

Postman API Documentation

Ledger & Co. REST API (Django 6.0 + DRF)

This document serves as the complete Postman API Reference Guide for the **Ledger & Co.** REST API (Django 6.0 + DRF).

A ready-to-import Postman Collection file is included in your project at:

```
c:\ecommerce_project_api\docs\postman_collection.json
```

1. Postman Setup & Variables

1.1 Collection Variables

Importing `docs/postman_collection.json` into Postman initializes these variables automatically:

Variable	Default Value	Description
<code>base_url</code>	<code>http://localhost:8000</code> (or <code>https://legible-unfrozen-elves.ngrok-free.dev</code>)	Root API server domain
<code>token</code>	(auto-saved by Login request)	DRF TokenAuthentication key
<code>product_id</code>	<code>1</code>	Test Product ID
<code>category_id</code>	<code>1</code>	Test Category ID
<code>order_id</code>	<code>1</code>	Test Order ID
<code>payment_id</code>	<code>1</code>	Test Payment ID

1.2 Authentication Header

For all protected endpoints, pass the header:

```
Authorization: Token {{token}}
```

Automated Token Capture: Running the `POST /api/auth/login/` request automatically extracts the returned token and populates `{{token}}` for all subsequent requests.

2. Complete API Reference

2.1 Authentication & User Domain (/api/auth/)

1. Register Account

- **HTTP Method:** `POST`
- **URL:** `{{base_url}}/api/auth/register/`
- **Auth:** None

Request Body (JSON):

```
{
  "username": "jane_doe",
  "email": "jane@example.com",
  "password": "StrongPass123!",
  "first_name": "Jane",
  "last_name": "Doe",
  "phone_number": "+8801700000000"
}
```

Response (201 Created):

```
{
  "id": 2,
  "username": "jane_doe",
  "email": "jane@example.com",
  "first_name": "Jane",
  "last_name": "Doe",
  "phone_number": "+8801700000000"
}
```

2. User Login

- **HTTP Method:** `POST`
- **URL:** `{{base_url}}/api/auth/login/`
- **Auth:** None

Request Body (JSON):

```
{
  "username": "jane_doe",
  "password": "StrongPass123!"
}
```

Response (200 OK):

```
{
  "token": "9944b09199c62bcf9418ad846dd0e4bbdfc6ee4b",
  "user": {
    "id": 2,
    "username": "jane_doe",
    "email": "jane@example.com",

```

```
    "first_name": "Jane",  
    "last_name": "Doe",  
    "is_staff": false  
  }  
}
```

3. Get User Profile

- **HTTP Method:** GET
- **URL:** `{{base_url}}/api/auth/profile/`
- **Auth:** Required (Token `{{token}}`)

4. Update Profile

- **HTTP Method:** PATCH
- **URL:** `{{base_url}}/api/auth/profile/`
- **Auth:** Required (Token `{{token}}`)

Request Body (JSON):

```
{  
  "first_name": "Janet"  
}
```

5. Logout Account

- **HTTP Method:** POST
- **URL:** `{{base_url}}/api/auth/logout/`
- **Auth:** Required (Token `{{token}}`)

2.2 Products & Categories Domain (/api/products/)

1. List & Search Products

- **HTTP Method:** GET
- **URL:** `{{base_url}}/api/products/?q=&category=&page=1`
- **Auth:** None

Query Parameters:

- **q:** Filter by name or SKU substring
- **category:** Filter by Category ID

Response (200 OK):

```
{
  "count": 15,
  "next": null,
  "previous": null,
  "results": [
    {
      "id": 1,
      "name": "Ergonomic Leather Chair",
      "sku": "SKU-CHAIR-01",
      "price": "299.99",
      "stock": 15,
      "status": "active",
      "category": 1,
      "is_available": true
    }
  ]
}
```

2. Get Product Detail

- **HTTP Method:** GET
- **URL:** `{{base_url}}/api/products/{{product_id}}/`
- **Auth:** None

3. Related Products (DFS Category Branch Recommendation)

- **HTTP Method:** GET
- **URL:** `{{base_url}}/api/products/{{product_id}}/related/`
- **Auth:** None

Description: Depth-First Search (DFS) traversal down the category subtree to recommend active, in-stock items.

4. Hierarchical Category Tree (DFS Traversal & Cached)

- **HTTP Method:** GET
- **URL:** `{{base_url}}/api/products/categories/tree/`
- **Auth:** None

Response (200 OK):

```
{
  "roots": [
    {
      "id": 1,
      "name": "Electronics",
      "slug": "electronics",
      "children": [
        {
          "id": 3,
          "name": "Laptops",
          "slug": "laptops",
          "children": []
        }
      ]
    }
  ]
}
```

5. Create Product (Staff / Admin)

- **HTTP Method:** **POST**
- **URL:** `{{base_url}}/api/products/`
- **Auth:** Staff User Token Required

Request Body (JSON):

```
{
  "name": "Wireless Mouse",
  "sku": "SKU-MOUSE-01",
  "description": "Ergonomic wireless mouse",
  "price": "19.99",
  "stock": 50,
  "status": "active",
  "category": 1
}
```

2.3 Orders Domain (/api/orders/)

1. Create Order

- **HTTP Method:** POST
- **URL:** {{base_url}}/api/orders/
- **Auth:** Required (Token {{token}})

Request Body (JSON):

```
{
  "items": [
    { "product_id": 1, "quantity": 2 }
  ]
}
```

Response (201 Created):

```
{
  "id": 10,
  "user": 2,
  "status": "pending",
  "total_amount": "599.98",
  "created_at": "2026-07-28T23:30:00Z",
  "items": [
    {
      "id": 15,
      "product": 1,
      "product_name": "Ergonomic Leather Chair",
      "quantity": 2,
      "price": "299.99",
      "subtotal": "599.98"
    }
  ]
}
```

2. List User Orders

- **HTTP Method:** GET
- **URL:** {{base_url}}/api/orders/
- **Auth:** Required (Token {{token}})

3. Get Order Detail

- **HTTP Method:** GET
- **URL:** {{base_url}}/api/orders/{{order_id}}/
- **Auth:** Required (Token {{token}})

2.4 Payments Domain (/api/payments/)

1. Initiate Payment (Stripe / bKash)

- **HTTP Method:** POST
- **URL:** {{base_url}}/api/payments/initiate/
- **Auth:** Required (Token {{token}})

Request Body (Stripe):

```
{
  "order_id": 10,
  "provider": "stripe"
}
```

Response (Stripe):

```
{
  "id": 5,
  "order": 10,
  "provider": "stripe",
  "transaction_id": "pi_3Mxxxxxx",
  "status": "pending",
  "amount": "599.98",
  "client_secret": "pi_3Mxxxxxx_secret_yyyyyy",
  "publishable_key": "pk_test_xxxxxx"
}
```

Request Body (bKash):

```
{
  "order_id": 10,
  "provider": "bkash"
}
```

Response (bKash):

```
{
  "id": 6,
  "order": 10,
  "provider": "bkash",
  "transaction_id": "TR0011xxxx",
  "status": "pending",
  "amount": "599.98",
  "bkash_url": "https://tokenized.sandbox.bka.sh/checkout?paymentID=TR0011xxxx"
}
```

2. Get Payment Detail

- **HTTP Method:** GET
- **URL:** {{base_url}}/api/payments/{{payment_id}}/
- **Auth:** Required (Token {{token}})

3. Confirm Payment (Gateway Confirmation & Atomic Stock Reduction)

- **HTTP Method:** POST
- **URL:** `{{base_url}}/api/payments/{{payment_id}}/confirm/`
- **Auth:** Required (Token `{{token}}`)

4. Query bKash Payment Status

- **HTTP Method:** GET
- **URL:** `{{base_url}}/api/payments/{{payment_id}}/query/`
- **Auth:** Required (Token `{{token}}`)

2.5 Payment Webhooks (/api/payments/webhooks/)

1. Stripe Webhook

- **HTTP Method:** POST
- **URL:** {{base_url}}/api/payments/webhooks/stripe/
- **Headers:** Stripe-Signature: t=..., v1=...

Request Body (JSON):

```
{
  "type": "payment_intent.succeeded",
  "data": {
    "object": {
      "id": "pi_3Mxxxxxx"
    }
  }
}
```

2. bKash Webhook & Callback Handler

- **HTTP Method:** GET or POST
- **URL:** {{base_url}}/api/payments/webhooks/bkash/?paymentID=TR0011xxxx&status=success
- **Aliases:** /payments/bkash/callback/
- **Auth:** None

Behavior:

- **JSON API Fetch:** Returns updated Payment object.
- **Browser Navigation:** Redirects browser directly to FRONTEND_URL + /order.html?id={orderID} upon successful payment confirmation.